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B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 3

COURSE: DATA STRUCTURES & ALGORITHMS (CS-301)

Time Allowed: 3 Hours Maximum Marks: 100 Marks Paper Code: CS-301

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Define Big-O, Omega, and Theta asymptotic notations with formal definitions.

Q1. (b) What is difference between linear and non-linear data structures?

Q1. (c) Explain LIFO and FIFO principles with practical computer science examples.

Q1. (d) What is priority queue? How is it implemented using max-heap?

Q1. (e) Define height balance factor in AVL tree. What are LL, RR, LR, RL rotations?

Q1. (f) Explain collision resolution techniques in hash tables: chaining vs open addressing.

Q1. (g) What is difference between BFS and DFS graph traversals?

Q1. (h) Define minimum spanning tree (MST) and state cycle property.

Q1. (i) What is worst case time complexity of Quick Sort? When does it occur?

Q1. (j) Differentiate between array and linked list based on memory and access time.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Convert the infix expression A + B * (C ^ D - E) / (F + G * H) into postfix using stack tabular trace.

Q2. (b) Write an algorithm to insert and delete elements from a Circular Queue using an array.

Q2. (c) Construct an AVL tree by inserting keys: 21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3 with rotation details.

Q2. (d) Find Minimum Spanning Tree using Prim's and Kruskal's algorithms for a given weighted graph.

Q2. (e) Explain Quick Sort algorithm with partitioning pseudocode and solve recurrence relation T(n) = 2T(n/2) + O(n).

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Write C functions to reverse a singly linked list and detect cycle using Floyd algorithm. [10 Marks]

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(b) Implement Doubly Linked List with insertion at beginning, middle, and deletion of a node. [10 Marks]

Q4. (a) Construct Binary Tree from Inorder: D B E A F C G and Preorder: A B D E C F G. [10 Marks]

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(b) Write algorithms for recursive and non-recursive Inorder tree traversal using explicit stack. [10 Marks]

Q5. (a) Apply Dijkstra single-source shortest path algorithm on a directed weighted graph with trace table. [10 Marks]

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(b) Explain Topological Sorting using Kahn BFS algorithm and DFS-based timestamping. [10 Marks]

DATA STRUCTURES & ALGORITHMS (CS-301) -- CONTINUED (PAGE 2)

Q6. (a) Explain Heap Sort with build-max-heap procedure and trace array: 45, 20, 80, 12, 60, 35, 90. [10 Marks]

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(b) Explain Merge Sort algorithm, write divide-and-conquer recurrence, and analyze space/time complexity. [10 Marks]

Q7. (a) Explain B-Tree of order m. Insert keys: 10, 20, 5, 6, 12, 30, 7, 17 into B-tree of order 3. [10 Marks]

-- OR --

(b) Discuss Hash Functions: Division, Mid-Square, Multiplication, and quadratic probing. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****